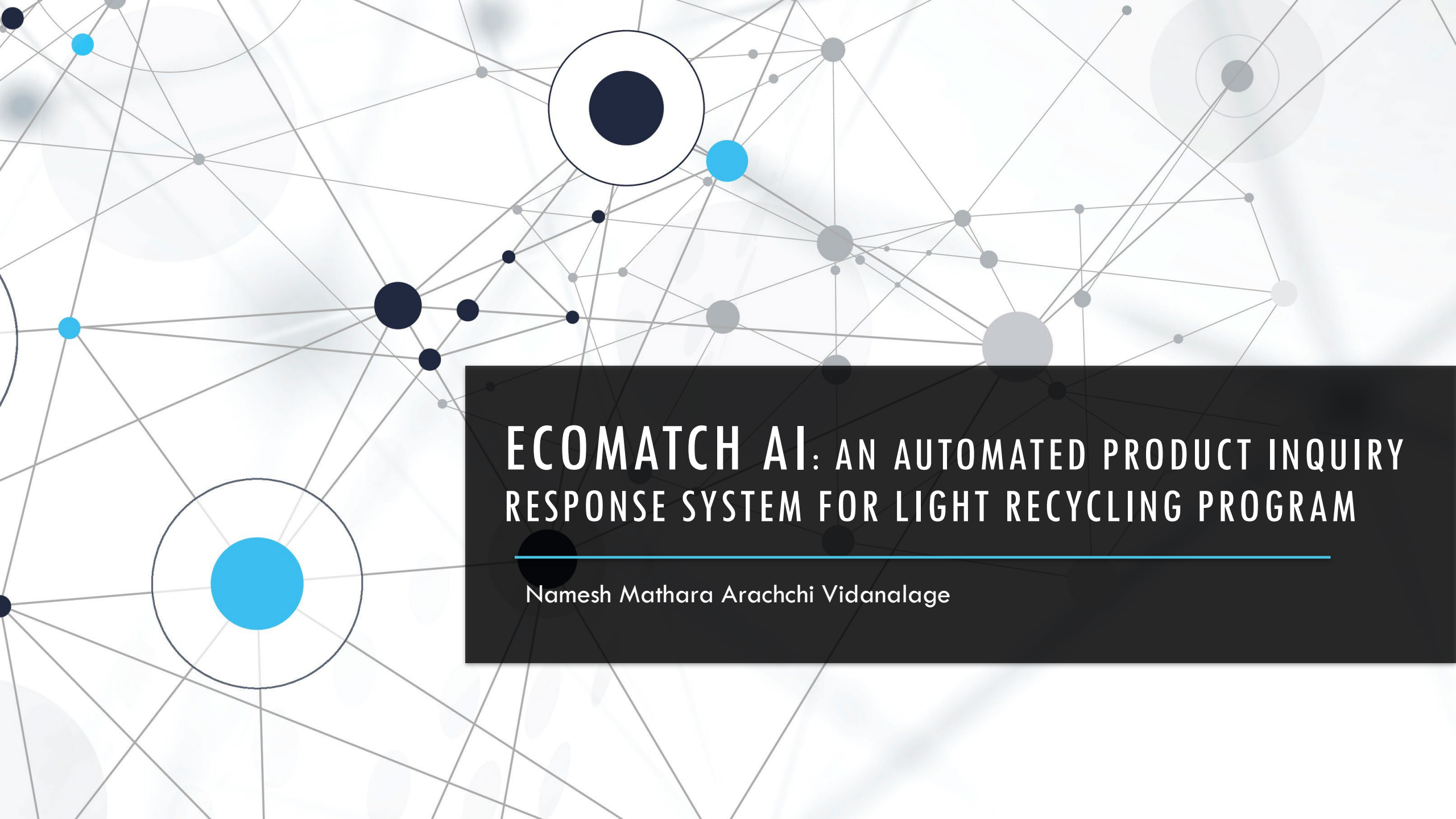
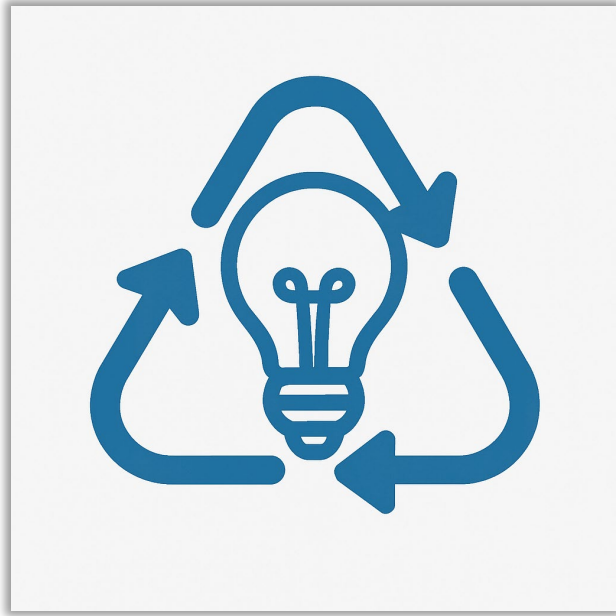


The background features a complex network diagram with numerous nodes of varying sizes (dark blue, light blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger concentric circles. The overall aesthetic is modern and technological.

ECOMATCH AI: AN AUTOMATED PRODUCT INQUIRY RESPONSE SYSTEM FOR LIGHT RECYCLING PROGRAM

Namesh Mathara Arachchi Vidanalage

BACKGROUND & PURPOSE



❑ **Product Care Association (PCA)**

- Recycling post-consumer lighting products across Canada

❑ **Manual product inquiry process**

- Inefficiencies and delays
- Inconsistent decision-making

RESEARCH QUESTION



❑ **How can I design and implement a product**

- ✓ End-to-end
- ✓ AI-powered
- ✓ Integrates text and image data
- ✓ Minimizes response times
- ✓ Enhances matching accuracy

NOVELTY



❑ **PCA's first ML-powered system**

Methodology and Tech Stack

Web Portal



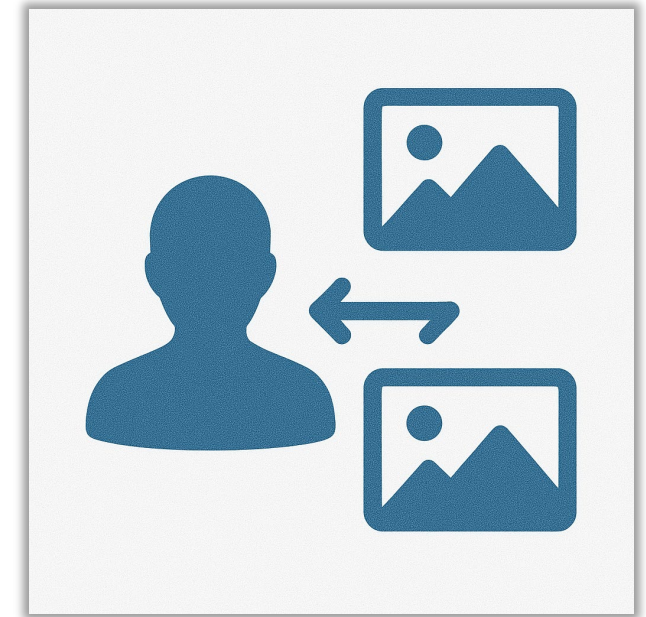
- ❑ **Frontend:** Python Flask and Bootstrap
- ❑ **Backend:** Python Flask
- ❑ **Data Storage:** Azure Table Storage and Azure Blob Storage

Text Similarity



- ❑ **Technique:** Sentence-BERT (SBERT) – fine-tuned transformer for semantic similarity
 - **Short text:** all-mpnet-base-v2
 - **Long text:** bert-large-nli-stsb-mean-tokens

Image Similarity



- ❑ **Technique:** Convolutional Neural Networks (CNN) based feature extraction for image similarity
 - **Best performer:** ResNet50

Challenges Faced



- ❑ CNN model - ResNet50 was slow, taking over 5 minutes to run
- ❑ Limited experience using Python in a full-stack development context
- ❑ Understanding and applying complex NLP and CNN models
- ❑ Insufficient labeled data for training ML models

Lessons Learned



- ❑ Precomputing embeddings drastically cut runtime from 5 minutes to 5 seconds, showing the value of optimizing for production
- ❑ Persistence and leveraging online resources made it possible to understand complex concepts
- ❑ Adding an admin review process improved data quality and helped generate valuable training data over time

Thank You!!!